

Python | Django | DRF | FastAPI | AI

Complete Interview Preparation Guide (with DSA)

425+ Questions & Answers | Topic-wise Organized

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Python Core & Advanced Concepts

91 Interview Questions & Answers

Q1. What is Python?

Python ek high-level, interpreted, object-oriented programming language hai. Ye simple syntax ki wajah se famous hai aur web development, AI, automation, scripting aur data science me use hoti hai. Features: Easy syntax, Readable code, Large community support, Cross-platform, Huge libraries support. Example: `print("Hello World")`

Q2. Python interpreted language q kehlati hai?

Python interpreted language is liye kehlati hai kyun ke iska code runtime par line by line execute hota hai. Python source code ko pehle bytecode me convert karta hai aur phir Python Virtual Machine (PVM) us bytecode ko execute karti hai.

Q3. Python interpreted hai ya compiled?

Python interpreted b hai or compiled b. Pehle step me code bytecode me compile hota hai, phir PVM (Python Virtual Machine) us bytecode ko line by line execute karti hai.

Q4. What are Data Types in Python?

Data Types different types ka data store karne ke liye use hote hain. Har variable ka ek data type hota hai jo batata hai ke us variable me kis type ka data store hoga. Categories: Numeric (int, float, complex), Sequence (list, tuple, range), Text (str), Set (set), Mapping (dict), Boolean (bool), Binary (bytes, bytearray, memoryview).

Q5. Variables kia hote hain?

Variable aik named memory location hota hai jo data ko temporarily store karta hai.

Q6. What is mutable and immutable in python?

Mutable wo data types hote hain jin ki value change ki ja sakti hai. Example: List, Dict, Set. Immutable wo data types hote hain jin ki value change nahi ki ja sakti. Example: Tuple, String, Int.

Q7. Difference between List and Tuple?

List mutable hoti hai, square brackets [] me define hoti hai, aur tuple se thori slow hoti hai. Tuple immutable hoti hai, parentheses () me define hoti hai, aur list se thori fast hoti hai.

Q8. What is Type Casting in Python?

Aik data type ko dusre data type me convert karna Type Casting kehlati hai. Do types hoti hain: Implicit - Python automatically convert karta hai. Explicit - Hum manually conversion karte hain.

Q9. Python me kitni type ke Operators hote hain?

7 Type: Arithmetic, Comparison, Logical, Assignment, Membership, Identity, Bitwise. Arithmetic: +, -, *, /, //, %, ** Comparison: ==, !=, >, <, >=, <= Logical: and, or, not

Q10. What is a function?

Function aik reusable block of code hota hai jo specific task perform karta hai. Example:

```
def greet(name):  
    return f"hello {name}"
```

Q11. Difference between parameter and argument?

Parameter wo variable hota hai jo function define karte waqt parentheses ke andar likha jata hai. Argument wo actual value hoti hai jo function call karte waqt pass ki jati hai.

Q12. Types of Arguments in Python?

Positional Arguments, Keyword Arguments, Default Arguments, Variable-Length Arguments (*args, **kwargs).

Q13. Positional arguments kia hote hain?

Jaha order (tarteeb) matter karti hai. Example:

```
def student(name, math, science):  
    print(f"{name} got {math}, {science}")  
student("Ali", 80, 90)
```

Q14. Keyword arguments kia hote hain?

Function call karte waqt parameter ka naam likh kar value pass ki jati hai. Example:

```
student(name="Ali", age=22)
```

Q15. Default arguments kia hote hain?

Jaha parameter ki value pehle se set hoti hai. Example:

```
def greet(name, message="Hello"):  
    print(f"{message}, {name}")
```

Q16. *args kya hai?

*args multiple positional arguments ko ek tuple ki tarah store karta hai, jab hume nahi pata hota ke function me kitni values ayengi.

Q17. **kwargs kya hai?

**kwargs multiple keyword arguments ko ek dictionary ki tarah receive karta hai.

Q18. What is the lambda function?

Lambda function aik anonymous (bina naam ka) function hota hai jo sirf ek expression rakhta hai. Example: square = lambda x: x**2

Q19. What is map()?

map() kisi function ko iterable ke har element par apply karta hai. Example: `list(map(lambda x: x*2, [1,2,3]))` -> [2,4,6]

Q20. What is filter()?

filter() sirf un elements ko return karta hai jo condition ko satisfy karte hain. Example: `list(filter(lambda x: x%2==0, [1,2,3,4]))` -> [2,4]

Q21. What is reduce()?

reduce() iterable ki tamam values ko combine karke ek single value return karta hai. Example: `from functools import reduce; reduce(lambda x,y: x*y, [1,2,3,4])` -> 24

Q22. What is scope in Python?

Scope variable ki accessibility (visibility) define karti hai. Python me 2 types ki scope hoti hai: Local Scope aur Global Scope. Global variable modify karne ke liye 'global' keyword use hota hai.

Q23. Common trap question: UnboundLocalError kab aata hai?

Jab function ke andar ek global variable ko bina 'global' keyword ke reassign kiya jaye lekin usay pehle print bhi kia jaye, to Python usay local variable treat kar leta hai aur UnboundLocalError deta hai.

Q24. What is recursion?

Recursion wo technique hai jisme function khud ko call karta hai. Ismein Base case (rukne ki condition) aur Recursive case (khud ko call karne ki condition) hote hain. Example: `factorial(n) = n * factorial(n-1)`

Q25. What is exception handling?

Exception Handling aik mechanism hai jo runtime errors ki wajah se program ko crash hone se bachata hai. try = risky code, except = error handle karo, finally = hamesha chalega, raise = khud error generate karo.

Q26. Difference between == and is?

== do objects ki value compare karta hai. is do objects ki memory location compare karta hai.

Q27. Deep copy vs shallow copy?

Deep copy poore object aur nested objects ki alag copy banati hai (original change ka koi asar nahi). Shallow copy naya object banati hai lekin nested objects ka reference share karti hai.

Q28. Iterable vs Iterator?

Iterable wo object hai jis par loop kiya ja sakta hai (`__iter__` provide karta hai) jaise list, tuple, string. Iterator wo object hai jo `next()` se ek ek value return karta hai.

Q29. What is a generator?

Generator aik special function hota hai jo yield keyword use karta hai aur values ko one by one generate karta hai, sari values ek sath memory me store nahi karta.

Q30. yield aur return me kya difference hai?

'return' function ko completely khatam kar deta hai. 'yield' function ko pause karta hai aur agli call par wahin se continue karta hai.

Q31. Generator kab use karte hain?

Large files read karne ke liye, large datasets process karne ke liye, memory optimization ke liye, streaming data ke liye.

Q32. Generator dobara use kyun nahi hota?

Generator ek one-time iterator hota hai. Values consume hone ke baad dobara use nahi hota, naya generator banana parta hai.

Q33. What is a decorator?

Decorator aik function hai jo kisi dusre function ki functionality ko bina change kiye modify ya extend karta hai. @ syntax se use hota hai. Common uses: Logging, Authentication, Timing, Caching, Validation.

Q34. Kitni types ke decorators hote hain?

Function decorators, Class decorators, aur Built-in decorators (@staticmethod, @classmethod, @property).

Q35. @property decorator kya karta hai?

@property kisi method ko attribute ki tarah access karne deta hai, bina parentheses ke call kiye.

Q36. What is Middleware in general terms?

Middleware wo component hota hai jo request aur response ke darmiyan kaam karta hai aur unhein process karta hai.

Q37. What is Synchronous Programming?

Synchronous Programming me code line by line execute hota hai. Ek task complete hone ke baad hi dusra task start hota hai.

Q38. What is Asynchronous Programming?

Asynchronous Programming me jab ek task wait (I/O operation) me ho to program dusra task start kar sakta hai. Ye application ko zyada efficient banata hai (event loop ke through concurrently).

Q39. async/await kya hote hain?

async keyword function ko coroutine banata hai aur await us coroutine ke complete hone ka wait karta hai bina program ko block kiye.

Q40. What is GIL in Python?

GIL (Global Interpreter Lock) Python ka ek lock hai jo ek waqt me sirf ek thread ko Python code execute karne deta hai. CPU-bound tasks me multithreading ka faida nahi hota, lekin I/O-bound tasks me multithreading effective hoti hai.

Q41. What is Multithreading?

Multithreading me ek process ke andar multiple threads hoti hain jo memory share karti hain. I/O-bound tasks (file, API, DB, network) ke liye best hai.

Q42. What is Multiprocessing?

Multiprocessing me multiple processes hoti hain, har process ki apni memory hoti hai aur alag CPU core use karti hai. CPU-bound tasks (image processing, ML) ke liye best hai.

Q43. Multithreading aur Multiprocessing me difference?

Multithreading same memory share karti hai aur I/O-bound tasks ke liye best hai. Multiprocessing alag memory rakhti hai aur CPU-bound tasks ke liye best hai. GIL multithreading ko limit karta hai, multiprocessing ko nahi.

Q44. Agar GIL ek hi thread chalne deta hai to Python multithreading ka faida?

I/O-bound tasks me jab ek thread wait kar rahi hoti hai to doosri thread kaam kar leti hai, is liye multithreading I/O tasks ke liye effective hai.

Q45. Kya GIL multiprocessing ko affect karta hai?

Nahi, GIL sirf ek process ke andar threads ko control karta hai. Har process ka apna GIL hota hai.

Q46. CPU-bound task kya hai?

CPU-bound task wo hota hai jo zyada tar CPU par heavy calculations karta hai, waiting kam hoti hai. Multiprocessing best hai.

Q47. I/O-bound task kya hai?

I/O-bound task wo hota hai jisme program zyada tar data read/write ya network response ka wait karta hai. Multithreading best hai.

Q48. What is Garbage Collection?

Garbage Collection aik automatic memory management process hai jo un objects ki memory free karta hai jo program me ab use nahi ho rahe.

Q49. Python me Garbage Collection kaise kaam karti hai?

Reference Counting (jab reference 0 ho jaye to memory free) aur Cyclic Garbage Collection (circular references detect karke memory free karti hai).

Q50. Python memory management ke liye kya use karta hai?

Reference Counting aur Garbage Collection.

Q51. What is a module in Python?

Module ek .py file hoti hai jisme functions, classes ya variables define hote hain aur usay import karke dusri files me use kia ja sakta hai.

Q52. What is a package in Python?

Package modules ka collection hota hai jo ek folder me hota hai aur usme `__init__.py` file hoti hai.

Q53. Difference between module and package?

Module ek single file hoti hai, jabke package multiple modules ka folder hota hai.

Q54. What are Python virtual environments?

Virtual environment ek isolated environment hota hai jisme project-specific packages install kiye jate hain taake different projects ke dependencies conflict na karein.

Q55. pip kya hai?

pip Python ka package manager hai jo libraries install, upgrade aur remove karne ke liye use hota hai.

Q56. What is PEP 8?

PEP 8 Python ki official style guide hai jo code ko readable aur consistent banane ke liye rules deti hai.

Q57. What is a docstring?

Docstring triple quotes me likha gaya documentation string hota hai jo function/class/module ka purpose describe karta hai.

Q58. List comprehension kya hai?

List comprehension ek concise tarika hai list banane ka. Example: `squares = [x**2 for x in range(5)]`

Q59. Dictionary comprehension kya hai?

Dictionary comprehension ek concise tarika hai dictionary banane ka. Example: `{x: x**2 for x in range(5)}`

Q60. What is slicing in Python?

Slicing sequence (list, string, tuple) ka ek portion nikalne ke liye use hota hai. Example: `a[1:4]`, `a[::-1]` (reverse)

Q61. Difference between `append()` and `extend()`?

`append()` ek single element list ke end me add karta hai. `extend()` ek iterable ke saare elements list me add karta hai.

Q62. Difference between `remove()`, `pop()` and `del`?

`remove()` value ke basis par element hataata hai. `pop()` index ke basis par element hataata hai aur return karta hai. `del` index ya poori list delete karne ke liye use hota hai.

Q63. What is a set in Python?

Set unordered collection hoti hai jisme duplicate values allowed nahi hoti.

Q64. Difference between set and frozenset?

Set mutable hota hai, frozenset immutable hota hai.

Q65. What are Python namespaces?

Namespace ek naming system hai jo har identifier (variable, function, class) ko unique naam deta hai taake naam clash na ho.

Q66. What is the difference between deep equality and shallow equality?

`==` operator value ki equality check karta hai (deep equality-ish for containers), `is` operator identity (memory location) check karta hai.

Q67. What are Python data classes?

@dataclass decorator classes banane ka boilerplate kam karta hai, jaise `__init__`, `__repr__` automatically generate ho jate hain.

Q68. What is context manager (with statement)?

Context manager resources (files, connections) ko automatically open aur close karne ke liye use hota hai. Example: `with open('file.txt') as f: data = f.read()`

Q69. `__enter__` aur `__exit__` methods kya karte hain?

Ye custom context manager banane ke liye use hote hain. `__enter__` resource setup karta hai, `__exit__` cleanup karta hai.

Q70. What are magic/dunder methods?

Magic methods Python ke special methods hote hain jin ke naam double underscores (`__`) se start aur end hote hain, jaise `__init__`, `__str__`, `__len__`, `__add__`.

Q71. `__str__` vs `__repr__`?

`__str__` user-friendly readable string return karta hai (print ke liye). `__repr__` developer-friendly unambiguous string return karta hai (debugging ke liye).

Q72. What is duck typing in Python?

Duck typing me object ka type important nahi hota, balke uske pass required method/attribute hona chahiye. "If it walks like a duck and quacks like a duck, it's a duck."

Q73. What is monkey patching?

Monkey patching runtime par kisi class ya module ke behavior ko dynamically change karna hota hai.

Q74. What is the difference between shallow function call and closures?

Closure wo function hai jo apne enclosing scope ki variables ko yaad rakhta hai chahe outer function khatam ho chuka ho.

Q75. What is a closure in Python?

Closure ek inner function hota hai jo outer function ki variables ko access kar sakta hai, chahe outer function ka execution khatam ho chuka ho. Example: `def outer(x):`

```
def inner():  
    return x  
    return inner
```

Q76. What is the difference between shallow copy of list using slicing and `copy()`?

Dono shallow copy hi karte hain lekin nested objects ka reference same rehta hai; dono me farq sirf syntax ka hai.

Q77. What is pickling and unpickling?

Pickling Python object ko byte stream me convert karna hai (serialization). Unpickling byte stream ko wapas Python object me convert karna hai (deserialization).

Q78. What is the difference between json and pickle?

json human-readable, language-independent format hai. pickle Python-specific binary format hai jo sab Python objects serialize kar sakta hai.

Q79. What is the walrus operator (`:=`)?

Walrus operator ek expression ke andar variable assign karne deta hai. Example: `if (n := len(data)) > 10: print(n)`

Q80. What are type hints in Python?

Type hints function parameters aur return values ke expected data type batane ke liye use hote hain, jaise `def add(a: int, b: int) -> int`.

Q81. What is the difference between shallow function scope and enclosing scope (LEGB rule)?

Python variable lookup LEGB rule follow karta hai: Local, Enclosing, Global, Built-in - is order me variable search hota hai.

Q82. What is the difference between is and == for immutable objects like small integers?

Python small integers (-5 to 256) ko cache karta hai is liye 'is' bhi True return kar sakta hai, lekin bade numbers ke liye alag memory locations hoti hain.

Q83. What is the difference between shallow copy and assignment (=)?

Assignment (=) sirf reference copy karta hai, dono variables same object point karte hain. Shallow copy naya object banata hai lekin nested references share karta hai.

Q84. What is unpacking in Python?

Unpacking se list/tuple ki values alag variables me assign ki ja sakti hain. Example: a, b, c = [1, 2, 3]

Q85. What is the zip() function?

zip() multiple iterables ko combine karke tuples ka ek iterator return karta hai. Example:
list(zip([1,2],[3,4])) -> [(1,3),(2,4)]

Q86. What is the enumerate() function?

enumerate() iterable ke elements ke sath unka index bhi return karta hai. Example: for i, val in enumerate(['a','b']): print(i, val)

Q87. What is the difference between range() and list?

range() ek memory-efficient sequence generator hai jo values on-demand generate karta hai. List saari values memory me store karti hai.

Q88. What is string formatting in Python (f-strings)?

f-strings variables ko directly string ke andar embed karne deta hai. Example: name = 'Ali';
print(f'Hello {name}')

Q89. What is exception chaining (raise from)?

raise ... from ... ek exception ko dusri exception ke context ke sath raise karta hai taake original error ka pata chal sake.

Q90. What is the difference between Error and Exception in Python?

Exception wo problem hai jo program handle kar sakta hai (try/except se). Error usually serious problem hota hai jise handle karna mushkil hota hai (jaise SyntaxError, MemoryError).

Q91. What is a custom exception?

Custom exception Exception class ko inherit karke banai jati hai taake specific error cases handle kiye ja sakein. Example: `class MyError(Exception): pass`

Python Object-Oriented Programming (OOP)

47 Interview Questions & Answers

Q92. What is OOP?

OOP (Object-Oriented Programming) aik programming paradigm hai jisme hum real-world entities ko Classes aur Objects ki form me represent karte hain.

Q93. Class vs Object?

Class aik blueprint (naqsha) hoti hai jiski basis par objects banaye jate hain. Object class ka real instance hota hai jo memory me create hota hai.

Q94. Instance aur Object me difference?

Practically dono terms ek hi meaning me use hoti hain. Har object class ka instance hota hai.

Q95. Kya ek class se multiple objects bana sakte hain?

Haan, ek class se multiple objects ya instances create kiye ja sakte hain.

Q96. Attributes kitni types ke hote hain?

2 types: Class Attributes (class ke andar, `__init__` ke bahar define hote hain) aur Instance Attributes (`__init__` ke andar `self` ke sath define hote hain).

Q97. Python me methods ki types kitni hoti hain?

3 types: Instance method (`self`), Class method (`@classmethod`, `cls`), Static method (`@staticmethod`).

Q98. Instance method kya hai?

Instance method `self` parameter leta hai aur object (instance) ke data ke sath kaam karta hai.

Q99. Class method kya hai?

`@classmethod` decorator use karta hai aur `cls` parameter leta hai. Ye class attributes ke sath kaam karta hai.

Q100. Static method kya hai?

`@staticmethod` decorator use karta hai. Ye na object ke data se related hota hai na class ke data se.

Q101. Self kya hota hai?

`self` current object (instance) ka reference hota hai.

Q102. What is constructor `__init__`?

Constructor `__init__()` Python ka special method hai jo object create hote hi automatically call hota hai. `__new__()` object create karta hai aur `__init__()` usay initialize karta hai.

Q103. What are Access Modifiers?

Access Modifiers control karte hain ke class ke data aur methods ko kon access kar sakta hai. 3 types: Public, Protected (`_var`), Private (`__var`).

Q104. Public access modifier kya hai?

Har jagah se access kiya ja sakta hai. Bina underscore ke likha jata hai, jaise `self.name`.

Q105. Protected access modifier kya hai?

Same class aur child class me use hota hai. Single underscore se likha jata hai, jaise `self._name`.

Q106. Private access modifier kya hai?

Sirf class ke andar use hota hai. Double underscore se likha jata hai, jaise `self.__name`. Name mangling ki wajah se bahir se direct access mushkil hota hai.

Q107. Name mangling kya hai?

Jab private attribute (`__name`) define kiya jata hai to Python usay internally `_ClassName__name` me convert kar deta hai taake accidental overriding na ho.

Q108. Association kia hai?

Jab aik class ka object doosri class ke object se related ho, to usay Association kehte hain.

Q109. Kya Association me dono objects independently exist kar sakte hain?

Haan, dono objects ek dusre ke bina bhi exist kar sakte hain.

Q110. Aggregation kia hai?

Aggregation Association ki special type hai jahan ek class doosri class ka object rakhti hai lekin dono independently exist kar sakte hain (HAS-A relationship).

Q111. Aggregation aur Association me difference?

Association sirf relationship batati hai. Aggregation me HAS-A relationship hoti hai aur ek class doosri class ka object rakhti hai.

Q112. Aggregation vs Composition?

Aggregation weak HAS-A relationship hai jahan child object parent ke baghair exist kar sakti hai. Composition strong HAS-A relationship hai jahan child object parent ke bina exist nahi kar sakta.

Q113. Four Pillars of OOP kya hain?

Encapsulation, Inheritance, Polymorphism, Abstraction.

Q114. What is Encapsulation?

Encapsulation me data aur methods ko ek class ke andar bundle kiya jata hai aur data ko direct access se protect kiya jata hai.

Q115. Data aur methods ko secure kyun kiya jata hai?

Taake unauthorized access na ho aur data safe rahe.

Q116. Getter aur Setter kya hain?

Getter/Setter Encapsulation ke parts hain jo private data ko access ya modify karne ke liye use hote hain. Getter value read karta hai, Setter value update karta hai.

Q117. What is Inheritance?

Inheritance me child class parent class ke methods ya properties ko inherit karti hai.

Q118. Types of Inheritance?

Single, Multi-level, Multiple, Hierarchical Inheritance.

Q119. Single Inheritance kya hai?

Jab ek child class ek parent class se inherit kare. Example:

```
class Animal:
    def eat(self): print('Eating')
class Dog(Animal): pass
```

Q120. Multiple Inheritance kya hai?

Jab ek child class multiple parent classes se inherit kare. Example: class Child(Father, Mother):
pass

Q121. Multilevel Inheritance kya hai?

Jab child class parent class se aur parent class grandparent class se inherit kare.

Q122. Hierarchical Inheritance kya hai?

Jab multiple child classes ek hi parent class se inherit karein.

Q123. What is Polymorphism?

Polymorphism same method ya function different objects ke liye different behaviors provide karta hai.

Q124. Types of Polymorphism?

Method Overloading aur Method Overriding.

Q125. What is Method Overloading?

Same class me same method different parameters ke sath use hota hai. Python directly overloading support nahi karti, lekin default arguments, *args, **kwargs se achieve kia ja sakta hai.

Q126. What is Method Overriding?

Child class parent class ke method ko same name se dobara define karti hai aur apna behavior implement karti hai.

Q127. What is Abstraction?

Sirf important cheezein dikhana aur implementation details hide karna Abstraction kehlata hai.

Q128. What is an abstract class in Python?

Abstract class ek aisi class hai jise directly instantiate nahi kia ja sakta, ismein abstract methods hote hain jo child class ko implement karne padte hain. ABC module se banayi jati hai.

Q129. What is an Interface concept in Python?

Python me formal interface nahi hota, lekin abstract classes ya protocols se interface jaisa behavior implement kia jata hai.

Q130. super() kia hai?

super() parent class ke methods ya constructor ko child class se call karne ke liye use hota hai.

Q131. Diamond problem kia hai?

Diamond Problem multiple inheritance me tab hota hai jab ek class do aisi classes se inherit kare jo dono ek common parent class ko inherit karti hon, is se method resolution me ambiguity paida hoti hai.

Q132. MRO kia hai?

MRO (Method Resolution Order) define karta hai ke Python pehle kis class ke method ko call karega jab multiple classes me same method ho.

Q133. What is Composition over Inheritance principle?

Ye principle kehta hai ke jahan tak mumkin ho, code reuse ke liye Inheritance ke bajaye Composition use ki jani chahiye, kyun ke composition zyada flexible aur loosely coupled hoti hai.

Q134. What is method resolution using C3 linearization?

Python MRO C3 linearization algorithm use karta hai jo consistent aur predictable order me classes ko resolve karta hai, especially multiple inheritance me.

Q135. What are SOLID principles (brief)?

Single Responsibility, Open/Closed, Liskov Substitution, Interface Segregation, Dependency Inversion - ye 5 principles clean aur maintainable OOP code likhne ke liye use hote hain.

Q136. What is operator overloading?

Operator overloading me hum dunder methods (`__add__`, `__sub__` etc.) use karke built-in operators ka custom behavior define karte hain classes ke liye.

Q137. What is class variable vs instance variable conflict?

Agar instance attribute ka naam class attribute se same ho to instance attribute priority leta hai us specific object ke liye.

Q138. What is a mixin class?

Mixin ek chhoti class hoti hai jo specific functionality provide karti hai aur multiple inheritance ke through dusri classes me add ki jati hai, bina independent use ke.

Data Structures & Algorithms (DSA)

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Q139. What is Data Structure?

Data Structure ek tareeqa hai data ko organize aur store karne ka taake usay efficiently access aur modify kiya ja sake.

Q140. What is an Algorithm?

Algorithm steps ka ek finite set hai jo kisi problem ko solve karne ke liye follow kiya jata hai.

Q141. Time Complexity kya hoti hai?

Time Complexity batati hai ke input size (n) barhne par algorithm ko complete hone ke liye kitne operations perform karne parenge.

Q142. Space Complexity kya hai?

Space Complexity batati hai ke algorithm ko run hone ke liye kitni extra memory chahiye hoti hai input size ke hisaab se.

Q143. Python/Django mein kaun si complexities sabse important hain?

$O(1)$ Constant - Dictionary lookup, List indexing. $O(\log n)$ Logarithmic - Binary Search. $O(n)$ Linear - Single loop. $O(n \log n)$ - Efficient sorting (Merge/Quick Sort avg). $O(n^2)$ Quadratic - Nested loops.

Q144. Best, Average aur Worst case complexity kya hoti hai?

Best case sabse kam operations wala scenario hota hai. Average case typical scenario hota hai. Worst case sabse zyada operations wala scenario hota hai - algorithms usually worst case ke basis par judge kiye jate hain.

Q145. Big O notation kya hai?

Big O notation algorithm ki upper bound time/space complexity describe karta hai, yani worst case performance.

Q146. Array kya hai?

Array ek data structure hai jo same data type ki multiple values ko ek hi variable me contiguous memory me store karta hai. Har element ka index 0 se start hota hai.

Q147. Array ki time complexity kya hai (access, search, insert, delete)?

Access: $O(1)$, Search: $O(n)$, Insert/Delete (end): $O(1)$, Insert/Delete (beginning/middle): $O(n)$.

Q148. List aur Array mein kya difference hai?

List Python ka built-in data structure hai jo different data types store kar sakti hai. Array same data type ke elements store karti hai, isliye memory efficient aur numerical operations ke liye fast hoti hai.

Q149. Python mein array use hoti hai ya list?

Python mein zyada tar list use hoti hai kyun ke wo different data types store kar sakti hai. array module bhi available hai lekin usme sirf same data type ki values store hoti hain.

Q150. What is a Linked List?

Linked List ek linear data structure hai jisme elements (nodes) memory me contiguous nahi hote, har node apne data aur next node ke address (pointer) ko store karta hai.

Q151. Types of Linked List?

Singly Linked List (sirf next pointer), Doubly Linked List (next aur previous pointer), Circular Linked List (last node first node se connected hota hai).

Q152. Array vs Linked List?

Array me elements contiguous memory me hote hain, fast access ($O(1)$) hota hai lekin insertion/deletion slow hoti hai. Linked List me dynamic size hoti hai, insertion/deletion fast ($O(1)$) hoti hai lekin access slow ($O(n)$) hota hai.

Q153. Linked List ki time complexity kya hai?

Access: $O(n)$, Search: $O(n)$, Insert/Delete at beginning: $O(1)$, Insert/Delete at end: $O(n)$ (bina tail pointer ke).

Q154. What is a Stack?

Stack ek linear data structure hai jo LIFO (Last In First Out) principle follow karta hai. Operations: push (add), pop (remove), peek (top dekhna).

Q155. Stack ke real-world use cases kya hain?

Undo/Redo functionality, Browser back button, Function call stack (recursion), Expression evaluation (balanced parentheses).

Q156. What is a Queue?

Queue ek linear data structure hai jo FIFO (First In First Out) principle follow karta hai. Operations: enqueue (add), dequeue (remove).

Q157. Types of Queue?

Simple Queue, Circular Queue, Priority Queue, Deque (Double-ended Queue).

Q158. Stack vs Queue?

Stack LIFO follow karta hai (last element pehle nikalta hai). Queue FIFO follow karta hai (pehla element pehle nikalta hai).

Q159. Queue ke real-world use cases kya hain?

Task scheduling, Print queue, Handling requests in a web server, BFS traversal.

Q160. What is Recursion (DSA context)?

Recursion me function khud ko call karta hai chhote sub-problems solve karne ke liye. Ismein Base case (rukne ki condition) aur Recursive case hoti hai.

Q161. Recursion vs Iteration?

Recursion function calls use karta hai aur zyada memory (call stack) leta hai lekin code clean hota hai. Iteration loops use karta hai, kam memory leta hai aur usually fast hoti hai.

Q162. What is a Hash Table / Dictionary?

Hash Table ek data structure hai jo key-value pairs store karta hai aur hash function use karke $O(1)$ average time me lookup, insert aur delete karta hai.

Q163. Python me dictionary ki average time complexity kya hai?

Lookup, Insert, Delete: $O(1)$ average case, lekin worst case $O(n)$ (hash collisions ki wajah se, jo rare hota hai).

Q164. What is a hash collision?

Hash collision tab hoti hai jab do different keys ka hash value same ho jaye. Isay chaining ya open addressing techniques se resolve kiya jata hai.

Q165. What is a Tree data structure?

Tree ek hierarchical (non-linear) data structure hai jisme nodes parent-child relationship me connected hote hain, ek root node se start hota hai.

Q166. What is a Binary Tree?

Binary Tree wo tree hai jisme har node ke maximum 2 children (left aur right) hote hain.

Q167. Binary Search Tree (BST) kya hai?

BST ek binary tree hai jisme left child ki value parent se choti aur right child ki value parent se badi hoti hai. Isse search operation $O(\log n)$ me hoti hai (balanced tree me).

Q168. Tree traversal ki types kya hain?

Inorder (Left-Root-Right), Preorder (Root-Left-Right), Postorder (Left-Right-Root), aur Level Order (BFS se).

Q169. BST ki time complexity kya hai?

Average case: Search/Insert/Delete $O(\log n)$. Worst case (skewed tree): $O(n)$.

Q170. What is a balanced tree (AVL/Red-Black)?

Balanced tree wo BST hoti hai jisme har node ke left aur right subtree ki height ka difference control me rakha jata hai, taake worst case $O(n)$ na ho aur operations $O(\log n)$ me rahein.

Q171. What is a Heap?

Heap ek complete binary tree hai jo Heap Property follow karta hai. Min-Heap me parent child se chota hota hai, Max-Heap me parent child se bara hota hai. Priority Queue implement karne ke liye use hota hai.

Q172. What is a Graph?

Graph ek non-linear data structure hai jo nodes (vertices) aur unke beech connections (edges) par mushtamil hota hai.

Q173. Types of Graphs?

Directed Graph (edges ka direction hota hai), Undirected Graph, Weighted Graph (edges ka weight/cost hota hai), Cyclic aur Acyclic Graph.

Q174. Graph representation ke tareeqe kya hain?

Adjacency Matrix (2D array, $O(V^2)$ space) aur Adjacency List (har node ki list of neighbors, $O(V+E)$ space, zyada efficient sparse graphs ke liye).

Q175. BFS (Breadth First Search) kya hai?

BFS graph/tree ko level by level traverse karta hai, Queue data structure use karke. Shortest path (unweighted graph) find karne ke liye use hota hai.

Q176. DFS (Depth First Search) kya hai?

DFS graph/tree ko ek path par jitna deep ja sakta hai jata hai phir backtrack karta hai, Stack (ya recursion) use karke.

Q177. BFS vs DFS?

BFS Queue use karta hai aur level-wise explore karta hai, shortest path ke liye best hai. DFS Stack/recursion use karta hai aur ek path ko fully explore karta hai, memory kam use karta hai.

Q178. What is Sorting?

Sorting data ko ek particular order (ascending/descending) me arrange karne ka process hai.

Q179. Bubble Sort kya hai?

Bubble Sort adjacent elements ko compare karke unhe swap karta hai agar wrong order me hon. Time Complexity: $O(n^2)$.

Q180. Selection Sort kya hai?

Selection Sort har pass me minimum element dhoond kar usay sahi position par rakhta hai. Time Complexity: $O(n^2)$.

Q181. Insertion Sort kya hai?

Insertion Sort elements ko ek ek karke already sorted part me sahi jagah insert karta hai. Time Complexity: $O(n^2)$, chhoti/nearly-sorted lists ke liye efficient hai.

Q182. Merge Sort kya hai?

Merge Sort Divide and Conquer approach use karta hai - array ko divide karta hai, sort karta hai phir merge karta hai. Time Complexity: $O(n \log n)$, stable sorting algorithm hai.

Q183. Quick Sort kya hai?

Quick Sort ek pivot element choose karta hai aur array ko us se choti aur bari values me partition karta hai. Average Time Complexity: $O(n \log n)$, Worst case: $O(n^2)$.

Q184. Merge Sort vs Quick Sort?

Merge Sort stable hota hai aur worst case me bhi $O(n \log n)$ guarantee karta hai lekin extra space ($O(n)$) leta hai. Quick Sort in-place hota hai (kam space) lekin worst case $O(n^2)$ ho sakta hai.

Q185. What is a stable sorting algorithm?

Stable sorting algorithm equal elements ka relative order maintain rakhta hai jaisa original array me tha. Example: Merge Sort, Insertion Sort.

Q186. Linear Search kya hai?

Linear Search array ke har element ko ek ek karke check karta hai. Time Complexity: $O(n)$.

Q187. Binary Search kya hai?

Binary Search sorted array par kaam karta hai, har step me search space ko aadha kar deta hai. Time Complexity: $O(\log n)$.

Q188. Binary Search ki precondition kya hai?

Array sorted hona chahiye, kyun ke Binary Search compare karke decide karta hai left ya right half me search continue karni hai.

Q189. What is a Two Pointer technique?

Two Pointer technique me array/list ke dono taraf se ya alag positions se do pointers move karke problem solve kiya jata hai, jaise sorted array me pair sum dhoondna. Time Complexity $O(n)$ tak reduce ho jati hai.

Q190. What is Sliding Window technique?

Sliding Window technique subarray/substring problems me use hoti hai jahan ek fixed ya variable size window data par slide karti hai taake nested loops se bacha ja sake. Time Complexity $O(n)$ tak

reduce hoti hai.

Q191. What is Dynamic Programming (DP)?

DP ek technique hai jisme complex problem ko chhote overlapping sub-problems me tor kar solve kiya jata hai aur unke results ko store (memoize) kiya jata hai taake dobara calculate na karna pare.

Q192. Memoization vs Tabulation?

Memoization Top-Down approach hai (recursion + caching results). Tabulation Bottom-Up approach hai (iterative tareeqe se table fill karna).

Q193. What is the Fibonacci sequence problem in DP context?

Fibonacci ko naive recursion se solve karna $O(2^n)$ leta hai kyun ke same sub-problems dobara calculate hote hain. DP (memoization/tabulation) se ye $O(n)$ me solve hota hai.

Q194. What is Greedy Algorithm?

Greedy Algorithm har step par locally best (optimal) choice select karta hai is umeed me ke overall bhi best result mile. Example: Coin change (kuch cases me), Activity selection.

Q195. Greedy vs Dynamic Programming?

Greedy har step ka best immediate decision leta hai aur wapas nahi jata, hamesha global optimal guarantee nahi karta. DP saare possible choices consider karta hai aur guaranteed optimal solution deta hai, lekin zyada time/space leta hai.

Q196. What is Divide and Conquer?

Divide and Conquer problem ko chhote sub-problems me divide karta hai, unhe independently solve karta hai, phir results ko combine karta hai. Example: Merge Sort, Quick Sort, Binary Search.

Q197. What is Backtracking?

Backtracking ek algorithmic technique hai jo step by step solution build karti hai aur agar koi path invalid nikle to wapas (backtrack) chali jati hai. Example: N-Queens problem, Sudoku solver.

Q198. What is a Palindrome check problem?

Palindrome check karta hai ke string forward aur backward same padhti hai ya nahi. Two-pointer technique se $O(n)$ time me solve hota hai.

Q199. How to find duplicate elements in an array efficiently?

Hash Set/Dictionary use karke elements ko track kiya jata hai. Agar element pehle se set me mojud ho to duplicate hai. Time Complexity: $O(n)$.

Q200. How to reverse a linked list?

Iteratively teen pointers (prev, current, next) use karke har node ka next pointer reverse kiya jata hai. Time Complexity: $O(n)$.

Q201. What is the two-sum problem approach?

Ek Hash Map use karke har element ka complement (target - element) check kiya jata hai. Agar complement map me mile to pair mil gaya. Time Complexity: $O(n)$ (brute force $O(n^2)$ ke muqable).

Q202. What is the difference between Depth and Height of a tree?

Depth kisi specific node ki root se distance hoti hai. Height us node se sabse door leaf tak ki distance hoti hai. Tree ki height root ki height hoti hai.

Q203. What is a Trie data structure?

Trie (Prefix Tree) ek tree-based data structure hai jo strings ko efficiently store aur search karta hai, especially prefix-based searches (autocomplete) ke liye.

Q204. What is the difference between Array and Hash Map in terms of use case?

Array index-based ordered data ke liye best hai (jaise sequential data). Hash Map key-based fast lookup ke liye best hai jab keys unique aur meaningful hon (jaise IDs, names).

Q205. What are common Big-O complexities interviewers expect you to know?

$O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, $O(2^n)$, $O(n!)$ - inka basic idea aur example algorithms yaad hone chahiye.

Django Framework

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Q206. What is Django?

Django ek high-level Python web framework hai jo fast development aur clean, pragmatic design ke liye famous hai. Ye "batteries-included" approach follow karta hai (authentication, admin panel, ORM built-in).

Q207. Django Project aur App me kya difference hai?

Django Project poori application hoti hai, jabke Django App us application ka ek specific module ya feature hota hai. Ek project ke andar multiple apps ho sakte hain.

Q208. What is MVT in Django?

MVT (Model-View-Template) Django ka architectural pattern hai jo application ko 3 parts me divide karta hai: Model (data), View (logic), Template (UI).

Q209. What is MVC?

MVC (Model-View-Controller) ek software architectural pattern hai jo application ko Model, View aur Controller me divide karta hai.

Q210. Django me MVC nahi MVT kyun hai?

Django me alag Controller nahi hota, balke Django ka URL dispatcher aur framework Controller ka role perform karte hain. Django me View, Controller ki responsibility nibhata hai aur Template UI show karta hai.

Q211. Django Request Lifecycle kya hai?

Request Browser -> URL -> Middleware -> View -> Model -> View -> Template/JSON -> Middleware -> Browser ke flow se process hoti hai.

Q212. Middleware kya hai?

Middleware Django ka component hai jo Request aur Response ke darmiyan kaam karta hai aur request ko View tak aur response ko browser tak jane se pehle process karta hai.

Q213. Middleware kitni baar execute hota hai?

Middleware 2 baar execute hota hai: Request ke time aur Response ke time.

Q214. Django me Middleware kitni types ke hote hain?

2 types: Request Middleware (auth check, session, logging, security) aur Response Middleware (response modify, headers add, logging).

Q215. Custom Middleware bana sakte hain?

Haan, apni requirement ke mutabiq custom middleware bana kar settings.py ki MIDDLEWARE list me add kiya ja sakta hai.

Q216. Common Built-in Middleware kaha define hote hain?

settings.py ke MIDDLEWARE list me: SecurityMiddleware, SessionMiddleware, CommonMiddleware, CsrfViewMiddleware, AuthenticationMiddleware, MessageMiddleware, XFrameOptionsMiddleware.

Q217. Model kya hai?

Model Django ki ek Python class hoti hai jo database table ko represent karti hai. Model ke har attribute database me ek field (column) banta hai.

Q218. ORM kya hai?

ORM (Object Relational Mapper) Python objects ke through database ke sath kaam karne ka tareeqa hai, jis se SQL likhne ki bajaye Python code se database operations kiye jate hain.

Q219. Migration kya hai?

Migration Django ka database schema ko update karne ka tareeqa hai.

Q220. makemigrations aur migrate me kya difference hai?

makemigrations model changes ke hisaab se migration files create karta hai. migrate un files ko database par apply karta hai.

Q221. Django Relationships kya hain?

Django Relationships models ke darmiyan connection banane ke liye use hoti hain: ForeignKey (Many-to-One), OneToOneField (One-to-One), ManyToManyField (Many-to-Many).

Q222. CASCADE kya hai?

CASCADE ek on_delete behavior hai. Agar parent object delete ho jaye to us se related child objects bhi delete ho jate hain.

Q223. on_delete ke aur options kya hain?

PROTECT (delete rok deta hai), SET_NULL (field ko null kar deta hai), SET_DEFAULT (default value set karta hai), DO_NOTHING (kuch nahi karta).

Q224. related_name kya hai?

related_name reverse relationship ka naam define karta hai, jis se parent object se related child objects ko access kar sakte hain.

Q225. Meta class kya hai?

Meta class model ki extra settings define karne ke liye use hoti hai, jaise ordering, table name aur permissions.

Q226. QuerySet kya hai?

QuerySet database se retrieve hone wale objects ka collection hota hai. Ye lazy loading follow karta hai, yani jab tak zarurat na ho query execute nahi hoti.

Q227. all() vs filter()?

all() table ke tamam records return karta hai. filter() condition ke hisaab se records return karta hai.

Q228. filter() vs get()?

filter() multiple records (QuerySet) return kar sakta hai. get() sirf ek object return karta hai.

Q229. get() vs filter() (exceptions)?

get() agar object na mile to DoesNotExist exception deta hai, aur agar multiple mil jayein to MultipleObjectsReturned exception deta hai. filter() empty QuerySet return karta hai, exception nahi deta.

Q230. exclude() kya karta hai?

Jo records condition ko match karte hain unhein hata kar baqi records return karta hai.

Q231. first(), last(), exists(), distinct() kya karte hain?

first() pehla object return karta hai. last() aakhri object return karta hai. distinct() duplicate records remove karta hai. exists() check karta hai QuerySet me record mojud hai ya nahi (True/False).

Q232. annotate(), aggregate(), order_by() kya karte hain?

annotate() har object ke sath extra calculated field add karta hai. aggregate() पूरी QuerySet par calculation karke ek result return karta hai (Sum, Avg, Count). order_by() records ko sort karta hai.

Q233. Function Based View (FBV) kya hai?

FBV ek simple function hoti hai jo request receive karti hai aur response return karti hai.

Q234. Class Based View (CBV) kya hai?

CBV ek class hoti hai jo GET, POST jaise HTTP methods ko alag methods ki form me handle karti hai.

Q235. FBV vs CBV?

FBV simple, easy, choti applications ke liye better hai. CBV reusable, inheritance support karti hai, complex applications ke liye better hai.

Q236. Generic Views kya hain?

Generic Views Django ki built-in CBVs hain jo common operations (List, Create, Update, Delete) ko asaan banati hain: ListView, DetailView, CreateView, UpdateView, DeleteView.

Q237. Django Signals kya hain?

Signals Django ka ek mechanism hai jo decoupled applications ko ek dusre ke events ke baare me notify karta hai.

Q238. Decoupled ka matlab?

Apps directly ek dusre par depend nahi karti, signal ke through communication hoti hai.

Q239. Django mein commonly used signals kaun se hain?

pre_save, post_save, pre_delete, post_delete, m2m_changed, pre_migrate, post_migrate.

Q240. select_related() vs prefetch_related()?

select_related() ForeignKey/OneToOneField ke liye SQL JOIN use karke ek hi query me related data fetch karta hai. prefetch_related() ManyToMany/Reverse ForeignKey ke liye multiple queries chalata hai aur Python me data combine karta hai.

Q241. N+1 Query Problem kya hai?

Jab ek main query chalti hai aur har related object ke liye alag query chalti hai, to usay N+1 Query Problem kehte hain. Is se performance slow hoti hai, select_related()/prefetch_related() se solve hota hai.

Q242. Database Indexing kya hai?

Index database ka data structure hai jo search, filtering aur sorting ko fast banata hai. Zyada use hone wale fields par index lagaya jata hai.

Q243. Gunicorn kya hai?

Gunicorn ek Python WSGI server hai jo Django application ko run karta hai aur requests process karta hai.

Q244. Nginx kya hai?

Nginx ek Reverse Proxy Server hai jo static/media files serve karta hai, security provide karta hai aur load balancing karta hai.

Q245. collectstatic command kya karti hai?

Ye sab apps ki static files ko STATIC_ROOT folder me collect karti hai taake Nginx unhein serve kar sake.

Q246. CSRF kya hai?

CSRF (Cross-Site Request Forgery) ek attack hai jisme attacker authenticated user se fake request send karwata hai. Django CSRF Token se is se protect karta hai.

Q247. XSS kya hai?

XSS (Cross-Site Scripting) me attacker website me malicious JavaScript inject karta hai. Django auto escaping se is se protection deta hai.

Q248. SQL Injection kya hai?

SQL Injection me attacker malicious SQL query inject karta hai. Django ORM parameterized queries use karta hai, is liye generally SQL Injection se safe hota hai.

Q249. CORS kya hai?

CORS (Cross-Origin Resource Sharing) decide karta hai ke kaun se domains aapki API ko access kar sakte hain.

Q250. settings.py kya hota hai?

settings.py Django project ki configuration file hai jisme database, installed apps, middleware, static files aur security settings define hoti hain.

Q251. urls.py kya kaam karta hai?

urls.py URL patterns ko views ke sath map karta hai taake Django ko pata chale kis URL par kaunsi view call karni hai.

Q252. What is a Django template?

Template HTML file hoti hai jisme Django Template Language (DTL) use karke dynamic data display kia jata hai.

Q253. Static files vs Media files?

Static files CSS, JS, images hoti hain jo developer provide karta hai. Media files wo files hoti hain jo users upload karte hain (jaise profile pictures).

Q254. What is Django Admin Panel?

Django admin ek built-in interface hai jisse models ka data easily manage (CRUD) kiya ja sakta hai bina custom UI banaye.

Q255. What is a Form in Django?

Django Form user input ko validate aur clean karne ke liye use hota hai, HTML forms ko Python classes ke through handle karta hai.

Q256. ModelForm kya hai?

ModelForm Model ke basis par automatically form fields generate karta hai, jis se code kam likhna parta hai.

Q257. What is the difference between Session and Cookies?

Cookies client (browser) side par store hoti hain. Session server side par store hoti hai, sirf session ID cookie me client ko di jati hai.

Q258. What is caching in Django?

Caching frequently used data ko temporarily store karta hai taake bar bar database query na chalani pare, is se performance improve hoti hai.

Q259. Types of caching in Django?

Per-view caching, Template fragment caching, Low-level cache API, aur Whole-site caching.

Q260. What is a context processor?

Context processor ek function hai jo templates ko extra context data provide karta hai jo har request ke sath available hota hai, bina har view me manually pass kiye.

Q261. What is Django's ORM lazy evaluation?

QuerySet database se data tab tak fetch nahi karta jab tak usay actually evaluate na kiya jaye (jaise loop, list(), print()). Isay lazy evaluation kehte hain.

Q262. bulk_create() kya hai?

bulk_create() multiple objects ko ek hi query me database me insert karta hai, is se performance better hoti hai multiple individual saves ke muqable.

Q263. What is a custom manager in Django models?

Custom Manager ek class hoti hai jo Model.objects ke through custom query methods provide karti hai.

Q264. What is F() expression in Django?

F() expression database field ki value ko dusri field ke sath compare ya update karne ke liye use hota hai, bina Python me value load kiye.

Q265. What is Q() object in Django?

Q() object complex queries banane ke liye use hota hai jisme AND, OR, NOT conditions combine ki ja sakti hain.

Q266. What is the difference between values() and values_list()?

values() dictionaries ka QuerySet return karta hai. values_list() tuples ka QuerySet return karta hai.

Q267. What is Django's transaction.atomic()?

transaction.atomic() ek block of database operations ko atomic banata hai, yani agar koi operation fail ho to sab operations rollback ho jate hain.

Q268. What is a slug field in Django?

Slug field URL-friendly short label hota hai jo usually title se generate hota hai (jaise 'my-first-post').

Q269. What is the difference between abstract base class and multi-table inheritance in models?

Abstract base class sirf common fields provide karti hai, apni table nahi banati. Multi-table inheritance har model ke liye alag table banati hai aur OneToOne link se connect karti hai.

Q270. What is Django Channels?

Django Channels Django ko WebSockets aur asynchronous protocols support karne ki ability deta hai, real-time features (chat, notifications) ke liye.

Q271. What is Celery used for with Django?

Celery ek task queue hai jo Django me background/asynchronous tasks (emails, notifications, heavy processing) run karne ke liye use hoti hai, usually Redis ya RabbitMQ ke sath.

Q272. What is the difference between `get_object_or_404` and `get()`?

`get_object_or_404()` object na milne par 404 error return karta hai. `get()` `DoesNotExist` exception raise karta hai jo manually handle karna parta hai.

Q273. What is Django's static vs staticfiles app?

'static' template tag static files ko URL me reference karta hai. 'staticfiles' app development aur production me static files ko manage/collect karne me madad karta hai.

Django REST Framework (DRF)

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Q274. What is Django REST Framework?

DRF Django ka framework hai jo RESTful APIs banane ke liye use hota hai. Isme serializers, authentication, permissions aur pagination jaise features built-in hote hain.

Q275. Serializer kya hota hai?

Serializer Python objects ko JSON me convert karta hai aur JSON data ko Python objects me convert karta hai.

Q276. ModelSerializer kya hai?

ModelSerializer automatically model ke basis par serializer aur fields generate karta hai. Is se code kam likhna parta hai.

Q277. Serializer vs ModelSerializer?

Serializer: fields manually define karni parti hain. ModelSerializer: fields model se automatically generate hoti hain.

Q278. APIView kya hai?

APIView DRF ki base class hai jo APIs banane ke liye use hoti hai aur GET, POST, PUT, DELETE requests handle karti hai.

Q279. GenericAPIView kya hai?

GenericAPIView APIView ka extended version hai jo serializer, queryset, pagination aur filtering ki support deta hai.

Q280. Mixins kya hain?

Mixins reusable classes hoti hain jo common functionality provide karti hain aur multiple views me reuse ki ja sakti hain: CreateModelMixin, ListModelMixin, UpdateModelMixin, DestroyModelMixin.

Q281. ViewSet kya hai?

ViewSet CRUD operations (Create, Read, Update, Delete) ko ek hi class me provide karta hai.

Q282. Router kya hai?

Router ViewSet ke URLs automatically generate karta hai.

Q283. APIView vs ViewSet?

APIView zyada custom control deta hai. ViewSet CRUD APIs ke liye kam code likhna parta hai aur Router ke sath use hota hai.

Q284. APIView ya ViewSet, CRUD ke liye kaunsa better hai?

ViewSet, kyun ke ye CRUD operations ko concise tareeqe se handle karta hai aur Router ke sath automatically URLs generate karta hai.

Q285. ModelViewSet kya hai?

ModelViewSet ek complete CRUD ViewSet hai jo automatically list, create, retrieve, update, destroy actions provide karta hai model ke basis par.

Q286. ReadOnlyModelViewSet kya hai?

ReadOnlyModelViewSet sirf list aur retrieve (GET) operations provide karta hai, create/update/delete allow nahi karta.

Q287. JWT kya hai?

JWT (JSON Web Token) stateless authentication method hai. User login ke baad token milta hai aur har request ke sath ye token bheja jata hai.

Q288. JWT ko Stateless kyun kehte hain?

Kyun ke server user ki session store nahi karta. User ki information token ke andar hoti hai.

Q289. JWT ke structure me kya hota hai?

JWT teen parts par mushtamil hota hai: Header, Payload, Signature - ye tino dot (.) se separate hote hain.

Q290. Access token vs Refresh token?

Access token short-lived hota hai aur API requests authenticate karta hai. Refresh token long-lived hota hai aur naya access token generate karne ke liye use hota hai.

Q291. Permissions kya hoti hain?

Permissions rules hoti hain jo decide karti hain ke kaun kisi API ya resource ko access kar sakta hai aur kya action perform kar sakta hai.

Q292. Common DRF Permissions kya hain?

AllowAny (kisi ko bhi access, login zaroori nahi), IsAuthenticated (sirf authenticated users), IsAdminUser (sirf admin users).

Q293. What is IsAuthenticatedOrReadOnly?

Ye permission unauthenticated users ko sirf read (GET) access deti hai, lekin write operations (POST, PUT, DELETE) ke liye authentication zaroori hoti hai.

Q294. Custom permission kaise banate hain?

BasePermission class ko inherit karke has_permission() aur has_object_permission() methods override kiye jate hain.

Q295. Throttling kya hai?

Throttling ek mechanism hai jo ek user ya client ki API requests ki speed aur quantity ko limit karta hai, taake server overload na ho aur API ka misuse na ho.

Q296. Agar user allowed limit se zyada requests bheje to?

Server uski additional requests ko temporarily block kar deta hai aur HTTP 429 (Too Many Requests) response return karta hai.

Q297. Throttling kyun use karte hain?

Taake koi user bohat zyada requests na bhej sake, server overload na ho, aur sab users ko fair aur secure access mil sake.

Q298. Types of throttling in DRF?

AnonRateThrottle (anonymous users ke liye), UserRateThrottle (authenticated users ke liye), ScopedRateThrottle (specific views ke liye).

Q299. Pagination kya hai?

Pagination large data ko multiple pages me divide karti hai taake performance aur response time better ho.

Q300. Types of pagination in DRF?

PageNumberPagination, LimitOffsetPagination, CursorPagination.

Q301. Validation kya hai?

Validation data save hone se pehle check karti hai ke data valid hai ya nahi.

Q302. DRF mein kitni types ki validation hoti hain?

3 types: Field-Level Validation (specific field, jaise email format), Object-Level Validation (multiple fields compare, jaise password==confirm password), Validators (built-in/custom, jaise unique email).

Q303. Field-level validation kaise likhte hain?

Serializer me `validate_<field_name>()` method define karke, jaise `validate_email(self, value)`.

Q304. Object-level validation kaise likhte hain?

Serializer me `validate(self, data)` method override karke multiple fields ko ek sath check kiya jata hai.

Q305. get() vs filter() (DRF context bhi)?

`get()` sirf ek object return karta hai (DoesNotExist/MultipleObjectsReturned exception de sakta hai).
`filter()` QuerySet return karta hai (empty ho sakta hai), exception nahi deta.

Q306. What is a Router's DefaultRouter?

DefaultRouter automatically CRUD URLs generate karta hai ViewSets ke liye aur ek API root view bhi provide karta hai jo available endpoints list karta hai.

Q307. What is content negotiation in DRF?

Content negotiation decide karta hai ke response kis format (JSON, XML) me client ko bheja jaye, based on Accept header.

Q308. What are Renderers in DRF?

Renderers decide karte hain ke API response kis format me convert hoga, jaise JSONRenderer, BrowsableAPIRenderer.

Q309. What are Parsers in DRF?

Parsers incoming request data ko parse karte hain (JSON, form data, multipart) taake view usay Python data structures ki tarah use kar sake.

Q310. What is HyperlinkedModelSerializer?

Ye ModelSerializer ki tarah hi hai lekin related objects ko primary key ki bajaye hyperlinks (URLs) ke through represent karta hai.

Q311. What is nested serializer in DRF?

Nested serializer ek serializer ke andar dusra serializer use karta hai taake related model ka data bhi nested JSON ki form me mile.

Q312. SerializerMethodField kya hai?

SerializerMethodField ek read-only field hai jiski value ek custom method (get_<field_name>) se calculate hoti hai.

Q313. What is DRF's browsable API?

Browsable API DRF ka built-in web-based interface hai jisse developer browser me directly API test kar sakta hai bina Postman use kiye.

Q314. What is versioning in DRF?

Versioning API ke different versions ko manage karne ka tareeqa hai, jaise URL versioning, Header versioning, Query parameter versioning.

Q315. What is the difference between authentication classes and permission classes?

Authentication classes decide karti hain user kon hai (identity verify). Permission classes decide karti hain us user ko kya access hai.

Q316. What is TokenAuthentication in DRF?

TokenAuthentication ek simple token-based authentication hai jisme user ko login ke baad ek static token milta hai jo har request ke header me bheja jata hai.

Q317. SessionAuthentication vs TokenAuthentication vs JWT?

SessionAuthentication cookies/session use karta hai (browser based). TokenAuthentication simple static token use karta hai. JWT stateless self-contained token use karta hai jisme expiry bhi hoti hai.

Q318. What is the purpose of to_representation() and to_internal_value() in serializers?

to_representation() Python object ko JSON/output format me convert karta hai. to_internal_value() incoming JSON data ko Python native data types me convert karta hai (validation se pehle).

Q319. What is DRF's ObtainAuthToken view?

Ye built-in view hai jo username/password lekar user ko authentication token return karta hai.

Q320. What is the use of format_suffix_patterns in DRF urls?

Ye URL patterns me .json, .api jaise suffixes add karne deta hai taake client specific format request kar sake.

FastAPI Framework

50 Interview Questions & Answers

Q321. What is FastAPI?

FastAPI ek modern, high-performance Python web framework hai jo APIs banane ke liye use hota hai. Ye Python type hints par based hai aur Starlette (web parts) aur Pydantic (data validation) par bana hota hai.

Q322. FastAPI itni fast kyun hai?

FastAPI ASGI (Asynchronous Server Gateway Interface) par based hai aur async/await support karta hai, jis se ye Django/Flask (WSGI-based) se kaafi zyada requests per second handle kar sakta hai.

Q323. FastAPI vs Django vs Flask?

Django full-featured, batteries-included framework hai (ORM, admin panel built-in). Flask lightweight, minimal micro-framework hai. FastAPI modern, async-first, high-performance API framework hai jo automatic validation aur documentation deta hai.

Q324. WSGI vs ASGI?

WSGI (Web Server Gateway Interface) synchronous hota hai, ek waqt me ek request handle karta hai. ASGI (Asynchronous Server Gateway Interface) asynchronous hota hai aur concurrently multiple requests handle kar sakta hai, WebSockets bhi support karta hai.

Q325. What is Pydantic?

Pydantic ek Python library hai jo data validation aur settings management ke liye use hoti hai. FastAPI ismein request/response data ko validate aur serialize karne ke liye use karta hai.

Q326. Pydantic BaseModel kya hai?

BaseModel Pydantic ki class hai jise inherit karke hum data schema define karte hain, jisme fields ke type hints diye jate hain aur Pydantic automatically validation karta hai.

Q327. FastAPI me path parameter kya hai?

Path parameter URL ke andar directly define hota hai, jaise `/items/{item_id}`, aur function parameter ke type hint se automatically validate hota hai.

Q328. FastAPI me query parameter kya hai?

Query parameter URL ke `'?'` ke baad `key=value` ki form me aata hai, jaise `/items?skip=0&limit=10`. Function parameters me default values de kar optional bhi banaya ja sakta hai.

Q329. Path parameter vs Query parameter?

Path parameter URL path ka hissa hota hai aur required hota hai (jaise /users/{id}). Query parameter URL ke end me optional filters ke liye hota hai (jaise ?page=1).

Q330. Request Body kya hai FastAPI me?

Request Body wo data hai jo client POST/PUT request ke sath JSON format me bhejta hai. Pydantic model se define kiya jata hai aur automatically parse/validate hota hai.

Q331. FastAPI me automatic documentation kaise generate hoti hai?

FastAPI OpenAPI standard follow karta hai aur automatically Swagger UI (/docs) aur ReDoc (/redoc) documentation generate karta hai, bina extra code likhe.

Q332. Swagger UI kya hai?

Swagger UI ek interactive web-based interface hai jo API endpoints ko test karne aur unki documentation dekhne ke liye use hota hai, FastAPI me /docs par automatically available hota hai.

Q333. Dependency Injection kya hai FastAPI me?

Dependency Injection ek design pattern hai jisme common logic (jaise database connection, authentication) ko 'Depends()' ke through reusable functions me define karke views me inject kiya jata hai.

Q334. Depends() kya karta hai?

Depends() FastAPI ka function hai jo dependency ko route function me inject karta hai, jaise authentication check, database session, ya common query parameters.

Q335. FastAPI me async def vs def route kab use karte hain?

async def tab use karte hain jab route ke andar I/O-bound operations (DB call, API call) async ho. Agar operations synchronous hain (blocking) to simple def use karte hain, FastAPI usay automatically threadpool me run karta hai.

Q336. What is a Path Operation in FastAPI?

Path Operation wo function hai jo kisi specific HTTP method (GET, POST, PUT, DELETE) aur URL path ke liye decorator (@app.get, @app.post) ke sath define hota hai.

Q337. FastAPI me response_model kya hota hai?

response_model parameter decorator me diya jata hai taake response ka data validate ho aur sirf specified fields hi client ko return hon (extra/sensitive fields exclude ho jayein).

Q338. Status codes FastAPI me kaise set karte hain?

status_code parameter decorator me set kiya jata hai, jaise @app.post('/items', status_code=201), ya HTTPException raise karke error status codes bhi return kiye ja sakte hain.

Q339. HTTPException kya hai?

HTTPException FastAPI ki class hai jo custom error response (status code + detail message) return karne ke liye use hoti hai, jaise `raise HTTPException(status_code=404, detail='Not Found')`.

Q340. FastAPI me Middleware kya hota hai?

Middleware FastAPI me har request/response ko process karne ke liye use hota hai (jaise CORS, logging, authentication) `@app.middleware('http')` decorator ke sath.

Q341. CORS Middleware FastAPI me kaise add karte hain?

CORSMiddleware ko `app.add_middleware()` ke through add kiya jata hai jisme allowed origins, methods aur headers specify kiye jate hain.

Q342. FastAPI me background tasks kya hain?

BackgroundTasks class use karke aise tasks (jaise email bhejna, logging) run kiye jate hain jo response return hone ke baad background me execute hote hain, client ko wait nahi karna parta.

Q343. FastAPI me database ke sath kaise kaam karte hain?

FastAPI khud ORM nahi deta, isliye SQLAlchemy ya Tortoise ORM jaise libraries ke sath integrate kiya jata hai, aur database session ko `Depends()` ke through inject kiya jata hai.

Q344. FastAPI me SQLAlchemy kaise use hoti hai?

SQLAlchemy models define karke, Session ko dependency ke through routes me inject kiya jata hai taake database operations (CRUD) perform ho sakein.

Q345. FastAPI me authentication kaise implement hoti hai?

OAuth2PasswordBearer, JWT tokens, ya API keys use karke authentication implement hoti hai. Security dependencies har protected route par `Depends()` ke sath lagayi jati hain.

Q346. OAuth2PasswordBearer kya hai?

Ye FastAPI ki built-in security utility hai jo token-based authentication (jaise JWT) implement karne me madad karti hai, login endpoint se token receive karta hai.

Q347. FastAPI me file upload kaise hota hai?

`UploadFile` aur `File()` ka use karke files upload kiye jate hain, jaise: `async def upload(file: UploadFile = File(...))`.

Q348. FastAPI me validation errors kaise handle hote hain?

Agar request data Pydantic model ke sath match na kare to FastAPI automatically 422 Unprocessable Entity response detailed error messages ke sath return karta hai.

Q349. Pydantic me Field() kya karta hai?

`Field()` extra validation rules aur metadata define karne ke liye use hota hai, jaise minimum/maximum length, default value, description.

Q350. Pydantic Validators kya hain?

@validator (Pydantic v1) ya @field_validator (v2) decorator use karke custom validation logic likhi ja sakti hai kisi field ke liye.

Q351. FastAPI me nested Pydantic models kya hote hain?

Nested models ek Pydantic model ke andar dusra Pydantic model use karte hain taake complex/related data structures represent ho sakein, jaise Address model Order model ke andar.

Q352. What is Starlette in context of FastAPI?

Starlette ek lightweight ASGI framework hai jispar FastAPI bana hota hai. Ye routing, middleware, aur WebSockets jaisi core web functionalities provide karta hai.

Q353. FastAPI me Uvicorn ka kya role hai?

Uvicorn ek ASGI server hai jo FastAPI application ko run karta hai, jaise: uvicorn main:app --reload.

Q354. FastAPI vs Flask performance ka farq kyun hota hai?

FastAPI async support ki wajah se concurrently zyada requests handle kar sakta hai, jabke Flask sync (WSGI) hone ki wajah se ek waqt me ek request process karta hai (bina extra setup ke).

Q355. FastAPI me Pydantic Settings kya hai?

BaseSettings class environment variables se app configuration (jaise DB URL, secret keys) automatically load karne ke liye use hoti hai.

Q356. FastAPI router kya hota hai?

APIRouter class use karke routes ko modules me organize kiya jata hai, jinhe baad me main app me include_router() se add kiya jata hai.

Q357. FastAPI me WebSockets kaise implement hote hain?

@app.websocket() decorator use karke real-time bidirectional communication (jaise chat, live notifications) implement ki jati hai.

Q358. FastAPI request validation aur Django validation me difference?

FastAPI Pydantic type hints se automatic validation karta hai (compile-time jaisa feel). Django Forms/Serializers ke through manual validation logic likhni parti hai.

Q359. FastAPI me exception handlers kaise customize karte hain?

@app.exception_handler() decorator use karke specific exceptions ke liye custom response format define kiya ja sakta hai.

Q360. FastAPI me testing kaise hoti hai?

TestClient (Starlette se) use karke, jo requests library jaisa interface deta hai taake endpoints ko bina server run kiye test kiya ja sake.

Q361. Query() aur Path() functions kya karte hain?

Ye FastAPI ke functions hain jo query/path parameters par extra validation (min_length, max_length, regex, description) add karne ke liye use hote hain.

Q362. What is data serialization in FastAPI?

Serialization Python objects (models) ko JSON me convert karne ka process hai jo response bhejte waqt automatically Pydantic ke through hota hai.

Q363. FastAPI me environment-based configuration kaise manage karte hain?

Pydantic BaseSettings ya python-dotenv ke through .env file se configuration load ki jati hai (dev/staging/production ke liye).

Q364. FastAPI ke real-world use cases kya hain?

Microservices, Machine Learning model serving (AI APIs), real-time applications, high-performance REST/GraphQL APIs.

Q365. FastAPI me GraphQL support hai?

Haan, FastAPI Strawberry ya Ariadne jaisi libraries ke sath integrate ho kar GraphQL APIs bhi support kar sakta hai.

Q366. What is the difference between FastAPI's request and response models?

Request model (Pydantic) incoming data validate karta hai. Response model outgoing data ko shape aur filter karta hai, sirf specified fields client ko return hote hain.

Q367. FastAPI me rate limiting kaise implement karte hain?

Third-party libraries jaise slowapi (FastAPI ke liye) ya custom middleware use karke request rate limit implement ki jati hai.

Q368. FastAPI me caching kaise implement karte hain?

fastapi-cache jaisi libraries ya Redis ke sath manually integrate karke response caching implement ki jati hai.

Q369. What is lifespan/startup and shutdown events in FastAPI?

@app.on_event('startup') aur @app.on_event('shutdown') (ya naya lifespan context manager) app start/stop hone par resources (jaise DB connection) initialize/cleanup karne ke liye use hote hain.

Q370. FastAPI me pagination kaise implement hoti hai?

Query parameters (skip, limit) manually define karke ya fastapi-pagination jaisi library use karke pagination implement ki jati hai.

Artificial Intelligence & Machine Learning

55 Interview Questions & Answers

Q371. What is AI (Artificial Intelligence)?

AI wo technology hai jisme machines ko insaano jaisi soch, learning aur decision-making ki ability di jati hai taake wo tasks (jaise recognition, prediction, reasoning) khud perform kar sakein.

Q372. AI, Machine Learning aur Deep Learning me difference?

AI ek broad concept hai jisme machines intelligent behavior dikhati hain. Machine Learning AI ka subset hai jisme machine data se seekh kar predictions karti hai. Deep Learning ML ka subset hai jo neural networks (multiple layers) use karta hai.

Q373. What is Machine Learning?

Machine Learning ek technique hai jisme algorithm data se patterns seekh kar bina explicitly programmed hue predictions ya decisions karta hai.

Q374. Types of Machine Learning?

Supervised Learning (labeled data se), Unsupervised Learning (unlabeled data se patterns dhoondna), Reinforcement Learning (reward/punishment se seekhna).

Q375. Supervised Learning kya hai?

Supervised Learning me model ko labeled data (input + correct output) diya jata hai aur wo input-output relationship seekhta hai. Example: Spam detection, Price prediction.

Q376. Unsupervised Learning kya hai?

Unsupervised Learning me model ko sirf input data diya jata hai (bina labels ke) aur wo khud patterns/groups dhoondta hai. Example: Customer segmentation (clustering).

Q377. Reinforcement Learning kya hai?

Reinforcement Learning me agent environment ke sath interact karke reward ya penalty ke basis par behavior seekhta hai. Example: Game playing AI, Robotics.

Q378. Classification vs Regression?

Classification me output ek category/class hoti hai (jaise spam/not spam). Regression me output ek continuous number hoti hai (jaise house price predict karna).

Q379. What is a training set, validation set and test set?

Training set model ko train karne ke liye use hota hai. Validation set model tuning/hyperparameters check karne ke liye. Test set final unbiased performance evaluate karne ke liye use hota hai.

Q380. What is Overfitting?

Overfitting tab hota hai jab model training data ko itna zyada seekh leta hai (noise samet) ke naye/unseen data par acchi performance nahi deta.

Q381. What is Underfitting?

Underfitting tab hota hai jab model itna simple hota hai ke wo training data ke patterns bhi sahi se nahi seekh pata, is se accuracy training aur testing dono par kam hoti hai.

Q382. Overfitting kaise rokte hain?

Regularization (L1/L2), more training data, cross-validation, dropout (neural networks me), aur model complexity kam karke.

Q383. What is Bias-Variance tradeoff?

Bias model ki simplifying assumptions ki wajah se error hai (underfitting). Variance model ki training data ke changes ke prati sensitivity hai (overfitting). Dono ke beech balance achieve karna zaroori hota hai.

Q384. What is Cross-Validation?

Cross-validation ek technique hai jisme data ko multiple folds me divide kiya jata hai aur model ko har fold par train/test karke performance ka average nikala jata hai (jaise K-Fold Cross Validation).

Q385. What is Feature Engineering?

Feature Engineering raw data se meaningful features (variables) create/select/transform karne ka process hai taake model ki performance behtar ho.

Q386. What is Feature Scaling / Normalization?

Feature Scaling data ki values ko ek specific range (jaise 0-1) me convert karti hai taake sab features equally model ko affect karein. Common methods: Min-Max Scaling, Standardization (Z-score).

Q387. Linear Regression kya hai?

Linear Regression ek supervised algorithm hai jo independent aur dependent variables ke beech linear relationship model karta hai, continuous values predict karne ke liye.

Q388. Logistic Regression kya hai?

Logistic Regression ek classification algorithm hai jo probability calculate karta hai (0-1 ke beech) sigmoid function use karke, usually binary classification ke liye.

Q389. Decision Tree kya hai?

Decision Tree ek tree-like model hai jisme data ko features ke basis par split kiya jata hai taake decisions/predictions ki jaa sakein, easily interpretable hota hai.

Q390. Random Forest kya hai?

Random Forest multiple Decision Trees ka ensemble hai jo unke predictions ko combine (voting/averaging) karke better accuracy aur less overfitting deta hai.

Q391. K-Nearest Neighbors (KNN) kya hai?

KNN ek simple algorithm hai jo naye data point ki classification/value uske 'k' nearest neighbors ke basis par decide karta hai.

Q392. Support Vector Machine (SVM) kya hai?

SVM ek classification algorithm hai jo data points ko classes me separate karne ke liye best possible hyperplane (decision boundary) dhoondta hai.

Q393. K-Means Clustering kya hai?

K-Means ek unsupervised algorithm hai jo data ko 'k' clusters me divide karta hai based on similarity, har cluster ka ek centroid hota hai.

Q394. What is a Neural Network?

Neural Network insaani dimagh ke neurons se inspire ek model hai jisme interconnected layers (input, hidden, output) hote hain jo data se complex patterns seekhte hain.

Q395. What is Deep Learning?

Deep Learning Machine Learning ka subset hai jo multiple hidden layers wale neural networks (deep networks) use karta hai complex tasks (image recognition, NLP) ke liye.

Q396. What is an Activation Function?

Activation Function neural network ke neuron ka output decide karti hai aur non-linearity introduce karti hai. Examples: ReLU, Sigmoid, Tanh, Softmax.

Q397. What is Backpropagation?

Backpropagation ek algorithm hai jo neural network ke weights ko error ke basis par update karta hai, gradient descent use karke, taake model ki accuracy improve ho.

Q398. What is Gradient Descent?

Gradient Descent ek optimization algorithm hai jo model ke loss function ko minimize karne ke liye parameters (weights) ko gradually update karta hai.

Q399. What is a Loss Function?

Loss Function model ki predictions aur actual values ke beech difference (error) measure karta hai. Examples: Mean Squared Error (regression), Cross-Entropy Loss (classification).

Q400. Epoch, Batch aur Iteration me difference?

Epoch pura training dataset ek baar model se guzarne ko kehte hain. Batch data ka chhota portion hota hai jo ek waqt me process hota hai. Iteration ek batch ke process hone ka count hai.

Q401. What is CNN (Convolutional Neural Network)?

CNN ek deep learning architecture hai jo mainly images process karne ke liye use hoti hai, convolution layers se features (edges, shapes) automatically extract karta hai.

Q402. What is RNN (Recurrent Neural Network)?

RNN ek neural network hai jo sequential/time-series data (jaise text, speech) process karne ke liye use hoti hai, kyun ke ye pichli information ko yaad rakhti hai.

Q403. What is NLP (Natural Language Processing)?

NLP AI ka wo field hai jo computers ko insaani zaban (text/speech) samajhne, process karne aur generate karne ki ability deta hai.

Q404. What is Tokenization in NLP?

Tokenization text ko chhote units (words, sub-words, ya characters - tokens) me todne ka process hai taake model usay process kar sake.

Q405. What are Embeddings?

Embeddings words ya sentences ko numerical vectors (numbers ki form) me convert karte hain taake unka semantic meaning (context) machine samajh sake.

Q406. What is a Transformer architecture?

Transformer ek deep learning architecture hai jo 'Attention Mechanism' use karta hai poore sequence ko parallel process karne ke liye, ye modern LLMs (GPT, BERT) ki base hai.

Q407. What is Attention Mechanism?

Attention Mechanism model ko decide karne me madad karta hai ke input sequence ke kaunse hisse par zyada focus karna hai kisi specific output generate karte waqt.

Q408. What is an LLM (Large Language Model)?

LLM ek bara neural network model hai (jaise GPT, Claude) jo bohat zyada text data par train hota hai aur human-like text samajhne aur generate karne me able hota hai.

Q409. What is Generative AI?

Generative AI wo AI systems hain jo naya content (text, images, audio, code) generate karte hain based on patterns jo unhone training data se seekhe hain.

Q410. What is Prompt Engineering?

Prompt Engineering ek technique hai jisme LLM se behtar aur accurate responses lene ke liye input (prompt) ko carefully design kiya jata hai.

Q411. What is Fine-tuning in AI?

Fine-tuning ek pre-trained model ko additional specific dataset par dobara train karne ka process hai taake wo particular task/domain me behtar perform kare.

Q412. What is Transfer Learning?

Transfer Learning me ek pre-trained model (jo already kisi bare dataset par train hua ho) ko naye related task ke liye reuse/adapt kiya jata hai, is se training time aur data ki zarurat kam hoti hai.

Q413. What is RAG (Retrieval-Augmented Generation)?

RAG ek technique hai jisme LLM ko response generate karte waqt external knowledge base/documents se relevant information retrieve karke context ke sath diya jata hai, taake accurate aur up-to-date answers milein.

Q414. What is a Vector Database?

Vector Database data ko embeddings (numerical vectors) ki form me store karta hai aur similarity search (semantic search) ke liye use hota hai, jaise Pinecone, Chroma, FAISS.

Q415. What is Hallucination in LLMs?

Hallucination tab hoti hai jab LLM confidently aisi information generate kar deta hai jo factually galat ya fabricated hoti hai.

Q416. What are common Python libraries used in AI/ML?

NumPy (numerical computing), Pandas (data manipulation), Scikit-learn (ML algorithms), TensorFlow aur PyTorch (deep learning), Matplotlib/Seaborn (visualization).

Q417. What is the difference between AI model inference and training?

Training wo phase hai jisme model data se seekhta hai (weights update hoti hain). Inference wo phase hai jisme trained model naye input par prediction/output deta hai.

Q418. What is Precision, Recall and F1-Score?

Precision batati hai ke positive predictions me se kitni sahi thin. Recall batata hai ke actual positives me se model ne kitne correctly identify kiye. F1-Score Precision aur Recall ka harmonic mean hai.

Q419. What is a Confusion Matrix?

Confusion Matrix ek table hai jo classification model ki performance dikhati hai: True Positive, True Negative, False Positive, False Negative ke through.

Q420. What is Accuracy in ML evaluation?

Accuracy total correct predictions ka percentage hai (correct predictions / total predictions), lekin imbalanced datasets me misleading ho sakti hai.

Q421. What is Data Preprocessing in ML?

Data Preprocessing raw data ko clean, transform aur organize karne ka process hai (missing values handle karna, encoding, scaling) taake model use kar sake.

Q422. What is One-Hot Encoding?

One-Hot Encoding categorical data ko binary vectors (0s aur 1s) me convert karta hai taake ML algorithms usay numerically process kar sakein.

Q423. How can AI models be integrated with a Django/FastAPI backend?

Trained ML/AI model ko pickle/joblib (ya ONNX) se save karke, ek API endpoint (Django view ya FastAPI route) banate hain jo input receive karke model se prediction return karta hai, aur ise as a microservice deploy kiya jata hai.

Q424. What is Model Deployment in AI?

Model Deployment trained ML model ko production environment me available karna hai taake real users/applications usay use kar sakein, usually API ke through (Flask/FastAPI/Django).

Q425. What is an API key and why is it used with AI services (like OpenAI/Anthropic)?

API key ek unique identifier hoti hai jo developer ko third-party AI service (jaise OpenAI, Anthropic) ki API access karne, usage track karne aur billing ke liye di jati hai.